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/ Practice Exam - 1

Correct
Answer

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Practice Exam - 1

Question 106 of 117



A 68-year-old man has been on femoral VA-ECMO for 8 hours for cardiogenic shock following a large anterior myocardial infarction. The bedside nurse notes the right leg (cannulated side) is cool and pale compared to the left, with diminished femoral and absent pedal pulses, and a weakened signal on continuous-wave Doppler at the ankle.

What is the most appropriate next step in management?

- A. Increase systemic heparin anticoagulation alone
-  B. Observe with serial limb examinations and reassess in 12 hours
- C. Increase ECMO pump flow to improve distal limb perfusion
- D. Proceed directly to above-knee amputation
-  E. Place an antegrade distal limb perfusion catheter to restore antegrade flow to the cannulated leg

Commentary

References

Content Code (Blueprint): Critical Care Cardiology — Mechanical Circulatory Support Complications (AHFTC Blueprint)

Task: Management

Teaching Point: Limb ischemia distal to the femoral arterial cannulation site is one of the most common vascular complications of peripheral VA-ECMO and is mitigated or treated with placement of a distal limb perfusion (reperfusion) catheter.

Correct Answer (Best Answer Choice): Place a distal limb perfusion catheter to restore antegrade flow to the cannulated leg

Distractors (Plausible Incorrect Options):

- Increase systemic heparin anticoagulation alone
- Observe with serial limb examinations and reassess in 12 hours
- Proceed directly to above-knee amputation
- Increase ECMO pump flow to improve distal limb perfusion

Rationale:

Correct answer E (distal perfusion catheter): The femoral arterial cannula can obstruct antegrade flow to the limb. A distal perfusion (reperfusion) catheter, placed antegrade or retrograde into the superficial femoral artery and connected to the arterial limb of the circuit, restores distal limb perfusion and is the standard treatment for cannula-associated limb ischemia. Risk factors for limb ischemia include female sex, smaller body surface area relative to cannula size, larger arterial cannula diameter, pre-existing peripheral arterial disease, diabetes mellitus, and prolonged duration of cannulation

Incorrect answers:

- A. Increasing anticoagulation alone: While adequate anticoagulation is important, it does not address the mechanical obstruction to flow caused by the cannula itself.
- B. Observation: Limb ischemia from arterial cannulation can progress rapidly to irreversible injury, compartment syndrome, or limb loss; watchful waiting in the setting of clinical ischemia is inappropriate.
- C. Increasing ECMO flow: Pump flow primarily affects the venous-to-arterial circuit flow and does not reliably improve antegrade perfusion distal to an obstructing arterial cannula.
- D. Immediate amputation: This is excessive in the absence of irreversible limb injury and is not the first-line management; restoration of perfusion should be attempted first.
- E. Correct answer